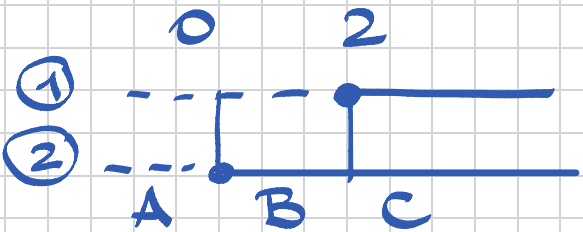
$$E = \{x \in \mathbb{R} : |x-2| + |x| - 3 \leq 0\}$$

① $x - 2 \geq 0 \qquad x \geq 2$

② $x \geq 0$



A) $\boxed{x \leq 0}$

$$2 - x - x - 3 \leq 0$$

$$-1 - 2x \leq 0 \qquad x \geq -\frac{1}{2}$$

$$\Rightarrow \boxed{-\frac{1}{2} \leq x \leq 0}$$

B) $\boxed{0 < x \leq 2}$

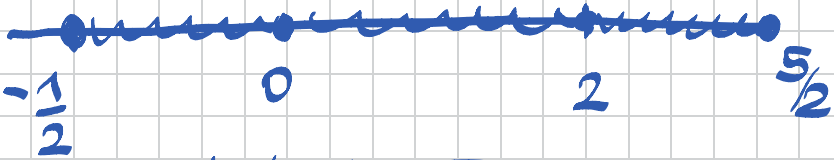
$$2 - x + x - 3 \leq 0 \quad \forall x$$

$$\Rightarrow \boxed{0 < x \leq 2}$$

c) $\boxed{x>2}$ $\quad x-2+x-3\leq 0$

$2x-5\leq 0 \quad x\leq 5/2$

$\boxed{2<x\leq 5/2}$



$$-\frac{1}{2}\leq x\leq \frac{5}{2}$$

$\min E = -\frac{1}{2}$ $\qquad$ $\max E = \frac{5}{2}$

$E=\{x\in R:\ 9^x+3^{x+1}-4\geq 0\}$

$9^x+3^{x+1}-4\geq 0$

$9^x+3^x\cdot 3-4\geq 0$ $\qquad 9^x=(3^x)^2$

$3^x=t \quad t^2+3t-4\geq 0$

$t_{1,2}=\frac{-3\pm 5}{2}=\begin{cases}-4\\1\end{cases}$

$t\leq -4 \vee t\geq 1$

$3^x \leq -4$ imp. $3^x \geq 1$ $x \geq 0$

$E = \{x \in \mathbb{R} : x \geq 0\}$

$\min E = 0$ $\sup E = +\infty$

$E = \{x \in \mathbb{R} : \log_2(\log_3(x^2-3x)) < 1\}$

$$\log_2(\log_3(x^2-3x)) < 1$$

CE $\begin{cases} x^2-3x>0 \\ \log_3(x^2-3x) > 0 \end{cases}$ $\begin{cases} x^2-3x>0 \\ x^2-3x>1 \end{cases}$

$x^2-3x-1>0$ $x_{1,2} = \frac{3 \pm \sqrt{13}}{2}$

CE: $x < \frac{3-\sqrt{13}}{2}$ v $x > \frac{3+\sqrt{13}}{2}$

$\log_3(x^2-3x) < 2$

$x^2-3x < 9$

$\log_a b = c$

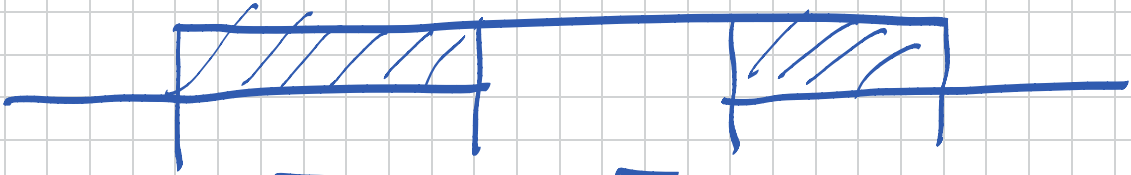
$b = a^c$

$x^2-3x-9<0$

$x_{1,2} = \frac{3 \pm \sqrt{45}}{2} = \frac{3 \pm 3\sqrt{5}}{2}$

$$\frac{3-3\sqrt{5}}{2} < x < \frac{3+3\sqrt{5}}{2}$$

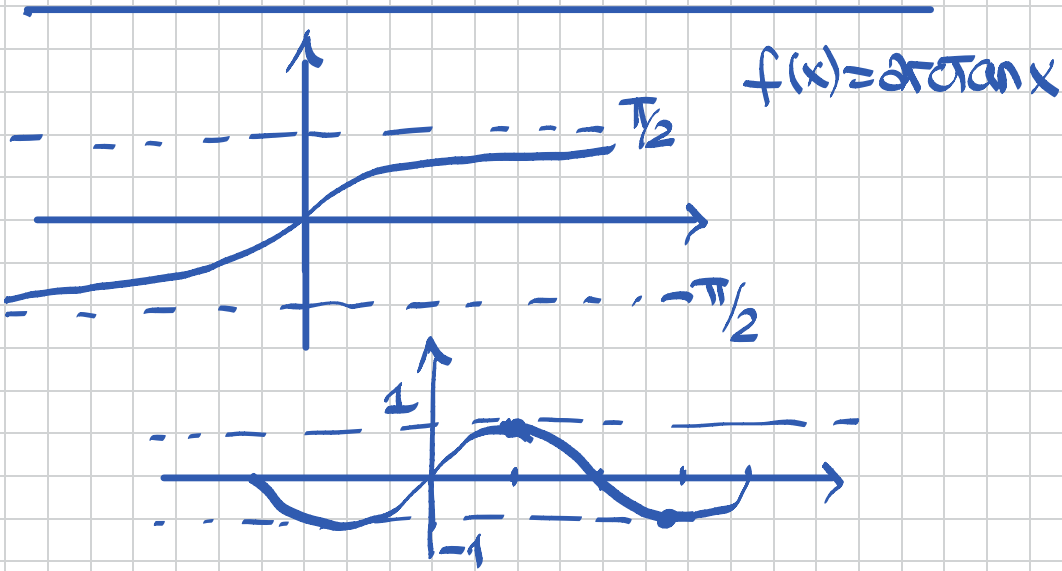
$$\frac{3-3\sqrt{5}}{2} \quad \frac{3-\sqrt{13}}{2} \quad \frac{3+\sqrt{13}}{2} \quad \frac{3+3\sqrt{5}}{2}$$



$$\Rightarrow \frac{3-3\sqrt{5}}{2} < x < \frac{3-\sqrt{13}}{2}$$

$$\frac{3+\sqrt{13}}{2} < x < \frac{3+3\sqrt{5}}{2}$$

$$\inf E = \frac{3-3\sqrt{5}}{2} \qquad \sup E = \frac{3+3\sqrt{5}}{2}$$



$z = a + ib \qquad a = \mathrm{Re}\, z \qquad b = \mathrm{Im}\, z$

$z = \rho(\cos\theta + i\sin\theta)$

1) $z = \dfrac{1+i}{3-i} + 1 = \dfrac{1+i+3-i}{3-i} =$

$= \dfrac{4}{3-i} \cdot \dfrac{(3+i)}{(3+i)} = \dfrac{12+4i}{9-i^2} = \dfrac{12+4i}{10}$

$= \dfrac{12}{10} + \dfrac{4}{10} i = \dfrac{6}{5} + \dfrac{2}{5} i \qquad \mathrm{Re}\, z = \dfrac{6}{5} \quad \mathrm{Im}\, z = \dfrac{2}{5}$

2) $z = \dfrac{1 + (1+3i)^2}{(1-i)^2 + i} = \dfrac{1+1+9i^2+6i}{1+i^2-2i+i}$

$= \dfrac{2-9+6i}{1-1-i} = \dfrac{-7+6i}{-i} \cdot \dfrac{i}{i}$

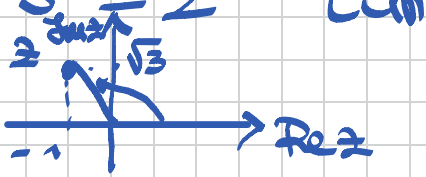
$= -\dfrac{(-7i-6)}{-1} = -6-7i$

3) $z = -\dfrac{1}{1+\sqrt{3}\,i} \qquad \rho = \sqrt{\mathrm{Re}^2 z + \mathrm{Im}^2 z}$

$\tan\theta = \dfrac{b}{a} = \dfrac{\mathrm{Im}\, z}{\mathrm{Re}\, z}$

$\rho = \sqrt{1+3} = 2 \qquad \tan\theta = -\sqrt{3}$

$\theta = \dfrac{2}{3}\pi$

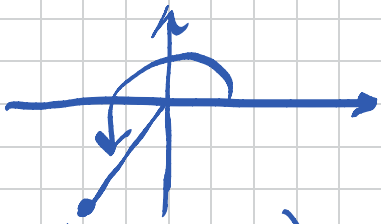


$z = 2\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right)$

4) $z = -2 - 2\sqrt{3}i$

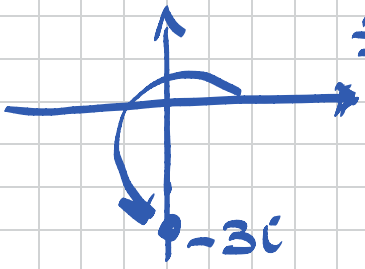
$\rho = \sqrt{4+12} = 4$

$\tan\theta = \frac{-2\sqrt{3}}{-2} = \sqrt{3} \qquad \theta = \frac{4}{3}\pi$



$z = 4\left(\cos\frac{4}{3}\pi + i\sin\frac{4}{3}\pi\right)$

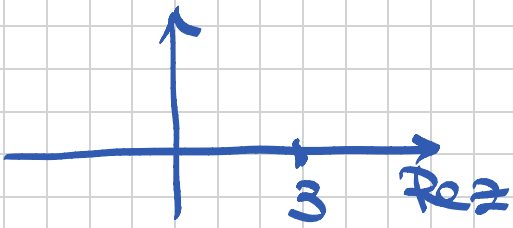
4) $z = -3i$ immaginario puro



$z = 3\left(\cos\frac{3}{2}\pi + i\sin\frac{3}{2}\pi\right)$

$-3i$

5) $z = 3$



$z = 3(\cos 0 + i\sin 0)$

$$\sqrt[n]{z} = \sqrt[n]{\rho}\left(\cos\frac{\theta+2k\pi}{n} + i\sin\frac{\theta+2k\pi}{n}\right)$$

$k = 0, 1, \dots n-1$

$\sqrt[5]{-4-4i}$ $\qquad z = -4-4i$

$\rho = \sqrt{16+16} = 4\sqrt{2} = \sqrt{32}$

$\tan\theta = 1 \qquad \theta = \frac{5\pi}{4}$

$$\sqrt[5]{z} = \sqrt[5]{\sqrt{32}}\left(\cos\frac{\frac{5\pi}{4}+2k\pi}{5} + i\sin\frac{\frac{5\pi}{4}+2k\pi}{5}\right)$$

$k = 0, 1, 2, 3, 4$

$k=0 \qquad w_0 = \sqrt[10]{2^5}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) =$

$\sqrt{2}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = 1+i$

$k=1 \qquad w_1 = \sqrt{2}\left(\cos\frac{\frac{5\pi}{4}+2\pi}{5} + i\sin\frac{\frac{5\pi}{4}+2\pi}{5}\right)$

$= \sqrt{2}\left(\cos\frac{13}{20}\pi + i\sin\frac{13}{20}\pi\right)$

$k=2 \qquad w_2 = \sqrt{2}\left(\cos\frac{\frac{5\pi}{4}+4\pi}{5} + i\sin\frac{\frac{5\pi}{4}+4\pi}{5}\right)$

$= \sqrt{2}\left(\cos\frac{21}{20}\pi + i\sin\frac{21}{20}\pi\right)$

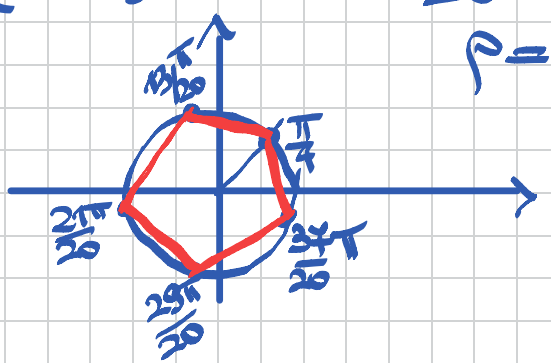
$k=3 \quad w_3 = \sqrt{2}\left(\cos\frac{4\cdot 5\pi+6\pi}{5} + i\sin\frac{4\cdot 5\pi+6\pi}{5}\right)$

$= \sqrt{2}\left(\cos\frac{26}{20}\pi + i\sin\frac{26}{20}\pi\right)$

$k=4 \quad w_4 = \sqrt{2}\left(\cos\frac{4\cdot 5\pi+8\pi}{5} + i\sin\frac{4\cdot 5\pi+8\pi}{5}\right)$

$= \sqrt{2}\left(\cos\frac{34}{20}\pi + i\sin\frac{34}{20}\pi\right)$

$\rho = \sqrt{2}$



$\sqrt[4]{-1-i\sqrt{3}}$

$\rho = 2$

$\tan\theta = \sqrt{3} \qquad \theta = \frac{4\pi}{3}$

$w_k = \sqrt[4]{z} = \sqrt[4]{2}\left(\cos\frac{\frac{4\pi}{3}+2k\pi}{4} + i\sin\frac{\frac{4\pi}{3}+2k\pi}{4}\right)$

$k=0,1,2,3$

$k=0 \quad w_0 = \sqrt[4]{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) =$

$= \sqrt[4]{2}\left(\frac{1}{2} + i\sqrt{3}/2\right)$

$k=1 \quad w_1 = \sqrt[4]{2}\left(\cos\frac{5}{6}\pi + i\sin\frac{5}{6}\pi\right) =$

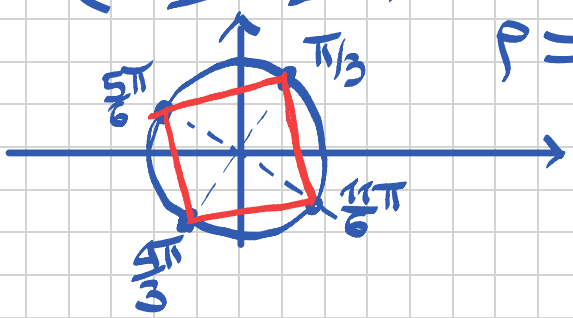
$= \sqrt[4]{2}\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

$k=2 \quad w_2 = \sqrt[4]{2}\left(\cos\frac{4}{3}\pi + i\sin\frac{4}{3}\pi\right) =$

$= \sqrt[4]{2}\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$

$k=3 \quad w_3 = \sqrt[4]{2}\left(\cos\frac{11}{6}\pi + i\sin\frac{11}{6}\pi\right)$

$= \sqrt[4]{2}\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$



$\rho = \sqrt[4]{2}$

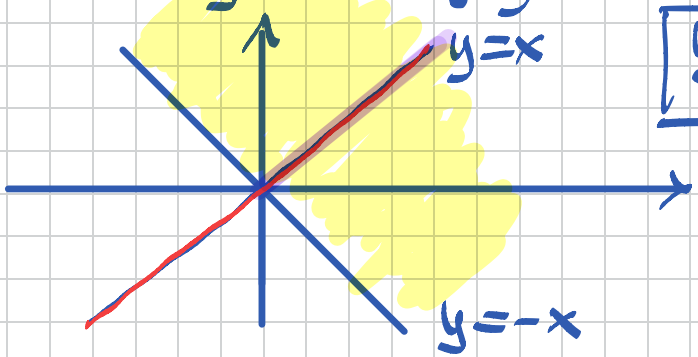
Determinare l'insieme dei numeri complessi che moltiplicati per $z = -3+3i$ diano un numero reale negativo e rappresentarli nel piano di Gauss

$w = x + iy$

$(-3+3i)(x+iy) = -3x - 3iy + 3ix +$

$-3y = -3x - 3y + i(-3y + 3x)$

$$\begin{cases} -3y+3x = 0 \\ -3x-3y < 0 \end{cases} \qquad \begin{cases} y = x \\ y > -x \end{cases}$$



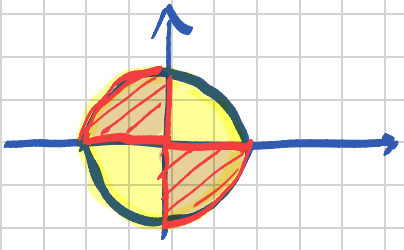
$y = x$, $x > 0$

2) Rappresentare nel piano di Gauss l'insieme dei numeri complessi che soddisfa il sistema

$$\begin{cases} \mathrm{Re}\, z \cdot \mathrm{Im}\, z \leq 0 \\ |z| \leq 10 \end{cases} \qquad \begin{cases} xy \leq 0 \\ \sqrt{x^2+y^2} \leq 10 \end{cases}$$

$$\begin{cases} xy \leq 0 \\ x^2+y^2 \leq 100 \end{cases}$$

$C(0,0)$ $r = 10$



$$A=\left\{z\in\mathbb{C}:\ \left|\bar{z}+\frac{6}{z}\right|\le 5\right\}$$

$$B=\{w\in\mathbb{C}:\ w=z+i\ \ z\in A\}$$

$$C=\{v\in\mathbb{C}:\ v=i\cdot w\ \ \ w\in B\}$$

$$A:\quad \left|\frac{\bar{z}z+6}{z}\right|\le 5 \qquad z\cdot\bar{z}=|z|^2$$

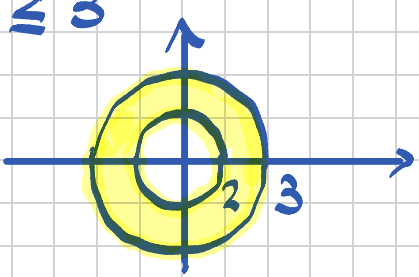
$$\left||z|^2+6\right|\le 5|z|$$

$$|z|^2+6-5|z|\le 0$$

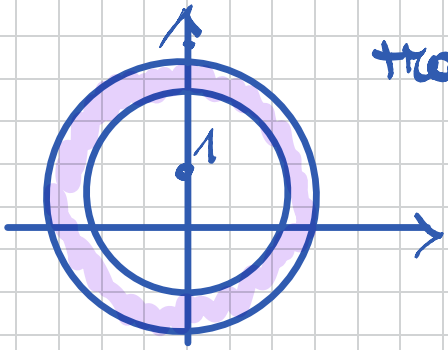
$$|z|^2-5|z|+6=0$$

$$|z|=\frac{5\pm 1}{2}=\begin{cases}3\\2\end{cases}$$

$$2\le|z|\le 3$$



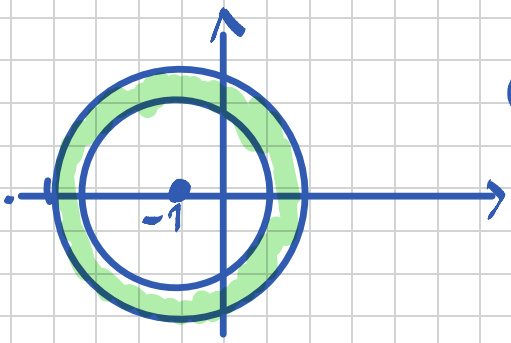
B:



traslazione verticale di A

$C(0,1)$ $r_1 = 2$ $r_2 = 3$

C: $i = \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$



rotazione di $\frac{\pi}{2}$ di B

$C(-1,0)$ $r_1 = 2$ $r_2 = 3$

EQUAZIONI IN $\mathbb{C}$

$z = x + iy$ $z \neq -3$

$$\frac{3z + 1 + 2i}{z + 3} = \bar{z}$$

$$\frac{3(x+iy) + 1 + 2i}{x + iy + 3} = x - iy$$

$$3x + 3iy + 1 + 2i = x^2 + xyi + 3x - xyi - i^2y^2 - 3iy$$

$$-x^2+6iy-y^2+1+2i=0$$

$$-x^2-y^2+1+i(+6y+2)=0$$

$$\begin{cases} -x^2-y^2+1=0 \\ 6y+2=0 \end{cases} \qquad \begin{cases} -x^2-\frac{1}{9}+1=0 \\ y=-\frac{1}{3} \end{cases}$$

$$\begin{cases} x^2=+8/9 \\ y=-\frac{1}{3} \end{cases} \qquad \begin{cases} x=\pm\frac{2\sqrt{2}}{3} \\ y=-\frac{1}{3} \end{cases}$$

$$\Rightarrow \quad z_1=\frac{2\sqrt{2}}{3}-\frac{1}{3}i \qquad z_2=-\frac{2\sqrt{2}}{3}-\frac{1}{3}i$$

$$2z+4i=\bar{z}\left(1+(\mathrm{Re}\,z)^2-\mathrm{Im}\,z\right)$$

$$2(x+iy)+4i=(x-iy)\left(1+x^2-y\right)$$

$$2x+2iy+4i=x+x^3-xy-iy-iyx^2+$$
$$+iy^2$$

$$x-x^3+xy+i\left(2y+4+y+yx^2-y^2\right)=0$$

$$\begin{cases} x-x^3+xy=0 \\ 3y+4+yx^2-y^2=0 \end{cases} \qquad \begin{cases} x(1-x^2+y)=0 \\ 3y+4+yx^2-y^2=0 \end{cases}$$

1) $\begin{cases} x=0 \\ 3y+4-y^2=0 \end{cases} \qquad y^2-3y-4=0$

$$\begin{cases} x=0 \\ y=4;\ y=-1 \end{cases} \qquad z_1 = 4i \quad z_2 = -1i$$

2) $$\begin{cases} 1-x^2+y=0 \\ 3y+4+yx^2-y^2=0 \end{cases}$$

$$\begin{cases} x^2 = 1+y \\ 3y+4+y+y^2-y^2=0 \end{cases} \qquad \begin{cases} x^2=0 \\ y=-1 \end{cases}$$

$z_3 = -i$

$$\Rightarrow z_1 = 4i,\ z_2 = z_3 = -i$$